Supervised Machine Learning

Week-8

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Example:

- A sample row from the dataset:
- $X = (Rainy, Hot, High, False),\ y = No$
- This represents:
- *What is the probability that someone will not play golf given that the weather is Rainy, Hot, High humidity, and No wind?*

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Predict if golf will be played (Yes or No).

- **Example Input:** $X = Sunny, Hot,\ Normal,\ False$.

Outlook

	Yes	No	P(yes)	P(no)
Sunny	3	2	3/10	2/4
Overcast	4	0	4/10	0/4
Rainy	3	2	3/10	2/4
Total	10	4	100%	100%

Humidity

	Yes	No	P(yes)	P(no)
Hot	3	4	3/9	4/5
Normal	6	1	6/9	1/5
Total	9	5	100%	100%

Temperature

	Yes	No	P(yes)	P(no)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
Total	9	5	100%	100%

Wind

	Yes	No	P(yes)	P(no)
False	6	2	6/9	2/5
True	3	3	3/9	3/5
Total	9	5	100%	100%

Play		P(Yes)/P(No)
Yes	9	9/14
No	5	5/14
Total	14	100%

For Class = Yes:

$$P(\text{Yes} \mid \text{today}) \propto \frac{2}{9} \cdot \frac{2}{9} \cdot \frac{6}{9} \cdot \frac{6}{9} \cdot \frac{9}{14}$$

$$P(\text{Yes} \mid \text{today}) \approx 0.02116$$

For Class = No:

$$P(\text{No} \mid \text{today}) \propto \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} \cdot \frac{2}{5} \cdot \frac{5}{14}$$

To compare:

$$P(\text{Yes} \mid \text{today}) = \frac{0.02116}{0.02116+0.0068} \approx 0.756$$

$$P(\text{No} \mid \text{today}) = \frac{0.0068}{0.02116+0.0068} \approx 0.244$$

Since:

$$P(\text{Yes} \mid \text{today}) > P(\text{No} \mid \text{today})$$

The model predicts: ***Yes (Play Golf)***

Naive Bayes Example: Text Classification

- Training Data

Message	Class	Total Word Count
"buy cheap meds"	Spam	3
"cheap offer today"	Spam	3
"let’s have lunch"	Not Spam	3
"project meeting today"	Not Spam	3

- Classes
- Spam (S)
- Not Spam (N)
- Calculate Prior Probabilities

$$P(S) = \frac{2}{4} = 0.5, \quad P(N) = \frac{2}{4} = 0.5$$

Naive Bayes Example: Text Classification

- Make vocabulory
- vocabulory size=10

- Laplace Smoothing
  - We add +1 so we don't get 0 when a word never appeared

$$P(word|class) = \frac{\text{count(word,class)} + 1}{\text{total words in class} + \text{vocabulary size}}$$

Word	In Spam	In Not Spam
buy	1	0
cheap	2	0
meds	1	0
offer	1	0
today	1	1
let's	0	1
have	0	1
lunch	0	1
project	0	1
meeting	0	1

- For word "cheap"

$$P(cheap|Spam) = \frac{2+1}{6+10} = \frac{3}{16} = 0.1875$$

$$P(cheap|NotSpam) = \frac{0+1}{6+10} = \frac{1}{16} = 0.0625$$

$$P(today|Spam) = \frac{1+1}{6+10} = \frac{2}{16} = 0.125$$

$$P(today|NotSpam) = \frac{1+1}{6+10} = \frac{2}{16} = 0.125$$

$$P(lunch|Spam) = \frac{0+1}{6+10} = \frac{1}{16} = 0.0625$$

$$P(lunch|NotSpam) = \frac{1+1}{6+10} = \frac{2}{16} = 0.125$$

- buy | 2/16 = 0.125 | 1/16 = 0.0625 |
- | meds | 2/16 = 0.125 | 1/16 = 0.0625 |
- | offer | 2/16 = 0.125 | 1/16 = 0.0625 |
- | meeting | 1/16 = 0.0625 | 2/16 = 0.125 |
- | project | 1/16 = 0.0625 | 2/16 = 0.125 |

$$P(S \mid M) = \frac{P(S)\,P(M \mid S)}{P(M)} \quad \text{where} \quad P(M) = P(S)\,P(M \mid S) + P(N)\,P(M \mid N)$$

$$P(M \mid S) = \frac{3}{16} \times \frac{1}{16} \times \frac{2}{16}.$$

$$P(M \mid N) = \frac{1}{16} \times \frac{2}{16} \times \frac{2}{16}$$

$$P(S \mid M) = \frac{3/4096}{5/4096} = \frac{3}{5} = 0.6, \qquad P(N \mid M) = \frac{2}{5} = 0.4.$$

classify "cheap lunch today" as Spam

Types of Naive Bayes Model

- **Gaussian Naive Bayes**, continuous values associated with each feature are assumed to be distributed according to a Gaussian distribution
- **Multinomial Naive Bayes**is used when features represent the frequency of terms (such as word counts) in a document. It is commonly applied in text classification, where term frequencies are important.
- **Bernoulli Naive Bayes** deals with binary features, where each feature indicates whether a word appears or not in a document. It is suited for scenarios where the presence or absence of terms is more relevant than their frequency. Both models are widely used in document classification tasks

Naïve bayes advantages

- Easy to implement and computationally efficient.
- Effective in cases with a large number of features.
- Performs well even with limited training data.
- It performs well in the presence of categorical features.
- For numerical features data is assumed to come from normal distributions

Naïve bayes disadvantages

- Assumes that features are independent, which may not always hold in real-world data.
- Can be influenced by irrelevant attributes.
- May assign zero probability to unseen events, leading to poor generalization.

Why Evaluate Models?

- Accuracy alone can be misleading
- Example: Spam detection – 95% non-spam → dumb model still gets 95% accuracy
- We need better evaluation metrics to understand performance

Model Evaluation

- Classification metrics measure how well a model predicts categorical labels.
- Confusion matrix is a simple table used to measure how well a classification model is performing. It compares the predictions made by the model with the actual results and shows where the model was right or wrong.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Confusion Matrix

- True Positive (TP): The model correctly predicted a positive outcome i.e the actual outcome was positive.
- True Negative (TN): The model correctly predicted a negative outcome i.e the actual outcome was negative.
- False Positive (FP): The model incorrectly predicted a positive outcome i.e the actual outcome was negative. It is also known as a Type I error.
- False Negative (FN): The model incorrectly predicted a negative outcome i.e the actual outcome was positive. It is also known as a Type II error.

Model Evaluation Metrics-Accuracy

- Accuracy shows how many predictions the model got right out of all the predictions.
- It gives idea of overall performance but it can be misleading when one class is more dominant over the other.
- For example, a model that predicts the majority class correctly most of the time might have high accuracy but still fail to capture important details about other classes. It can be calculated using the below formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Model Evaluation Metrics-Precision

- Precision focus on the quality of the model's positive predictions. It tells us how many of the "positive" predictions were actually correct. It is important in situations where false positives need to be minimized such as detecting spam emails or fraud. The formula of precision is:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Model Evaluation Metrics-Recall

- Recall measures how how good the model is at predicting positives. It shows the proportion of true positives detected out of all the actual positive instances. High recall is essential when missing positive cases has significant consequences like in medical tests.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Example

Activity: For the given confusion matrix, calculate accuracy, precision, recall, F1.

	Pred: Positive	Pred: Negative
Actual: Positive	TP = 40	FN = 20
Actual: Negative	FP = 10	TN = 30

Model Evaluation Metrics-F1 Score

- F1-score combines precision and recall into a single metric to balance their trade-off. It provides a better sense of a model’s overall performance particularly for imbalanced datasets.
- Imbalanced datasets-> meaning one class appears much more frequently than another.
- It is the harmonic mean of precision and recall which combine both metrics into a single value that balances their importance.

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

Example:

- Let's take an example of a dataset with 100 total cases. Out of these 90 are positive and 10 are negative cases. The model predicted 85 positive cases out of which 80 are actual positive and 5 are from actual negative cases. The confusion matrix would look like:

	Pred: Positive	**Pred: Negative**	**Total**	**Recall**
Actual: Positive	TP = 80	FN =5	85	80/80+5=85/90 =0.94
Actual: Negative	FP = 10	TN = 5	15	
Total	90	10		
Precision	80/80+10=80/90=0.88		ACCURACY= 80+5/100=85%	F1=0.91

Example:

- Consider the below case where there are only 9 cases of true positives out of a dataset of 100.

	Actual		Total
Model Prediction	1	1	2
	8	90	98
Total	9	91	100

- In this case if we give importance to accuracy over model will predict everything as negative. This gives us an accuracy of 91 %. However our F1 score is low.

- However one must also consider the opposite case where the positives outweigh the negative cases. In such a case our model will try to predict e

	Actual		Total
Model Prediction	90	8	98
	1	1	2
Total	91	9	100

- Here we get a good F1 score but low accuracy. In such cases the negative should be treated as positive and positive as negative.

F1 Score for Multi-Class Problem

- In a multi-class classification problem where there are more than two classes we calculate the F1 score per class rather than providing a single overall F1 score for the entire model. This approach is often referred to as the one-vs-rest (OvR) or one-vs-all (OvA) strategy.
- For each class in the multi-class problem a binary classification problem is created.We treat one class as the positive class and the rest of the classes as the negative class.
- Once we have calculated the F1 score for each class we aggregate these scores to get an overall performance measure for your model.
- Common approaches include calculating a micro-average, macro-average or weighted average of the individual F1 scores.

F1 Score for Multi-Class Problem

- Micro-average: Calculates metrics globally by counting the total true positives, false negatives and false positives.
- Macro-average: Averages the F1 score for each class without considering class imbalance.
- Weighted-average: Considers class imbalance by weighting the F1 scores by the number of true instances for each class.

ROC Curve & AUC

- A graph used to check how well a binary classification model works.
- It helps us to understand how well the model separates the positive cases like people with a disease from the negative cases like people without the disease at different threshold level.
- It shows how good the model is at telling the difference between the two classes by plotting: ROC Curve :
  - X-axis: False Positive Rate (FPR)
  - Y-axis: True Positive Rate (TPR) = Recall

$$\text{TPR (Recall)} = \frac{TP}{TP + FN}$$

$$\text{FPR} = \frac{FP}{FP + TN}$$

Balancing sensitivity and specificity

- True Positive Rate (TPR): how often the model correctly predicts the positive cases also known as Sensitivity or Recall.
- False Positive Rate (FPR): how often the model incorrectly predicts a negative case as positive.
- Specificity: measures the proportion of actual negatives that the model correctly identifies. It is calculated as 1 - FPR.

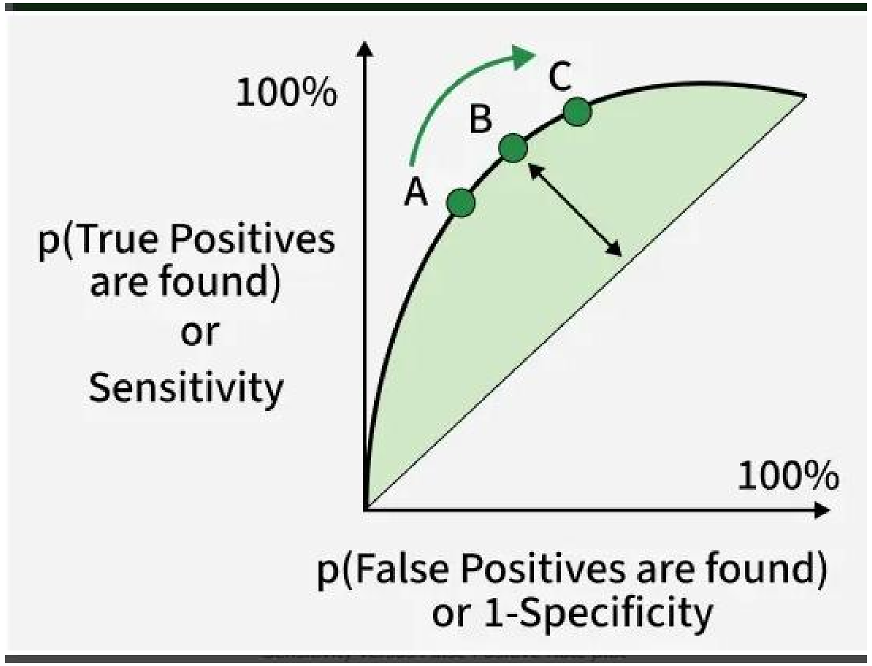
$$\mathrm{TPR\ (Recall)} = \frac{TP}{TP + FN}$$

$$\mathrm{FPR} = \frac{FP}{FP + TN}$$

ROC Curve & AUC

- The higher the curve the better the model is at making correct predictions.

Term	Meaning
TPR (Recall)	how often the model correctly predicts the positive cases also known as Sensitivity or Recall.
FPR	how often the model incorrectly predicts a negative case as positive.
ROC Curve	TPR vs FPR at different thresholds
AUC	Overall ability of model to separate classes-measures the area under the ROC curve.



- Let’s say we have **6 samples** and a model that outputs **probabilities of being positive**.
- Step 1 ☐ : Sort by Predicted Probability (highest → lowest)
- Step 2 ☐ : Choose different thresholds and compute (TPR, FPR)
- Step 3 ☐ : Plot ROC Curve Points

Sample	Actual Label	Predicted Probability
A	1	0.95
B	0	0.90
C	1	0.85
D	0	0.70
E	1	0.40
F	0	0.10

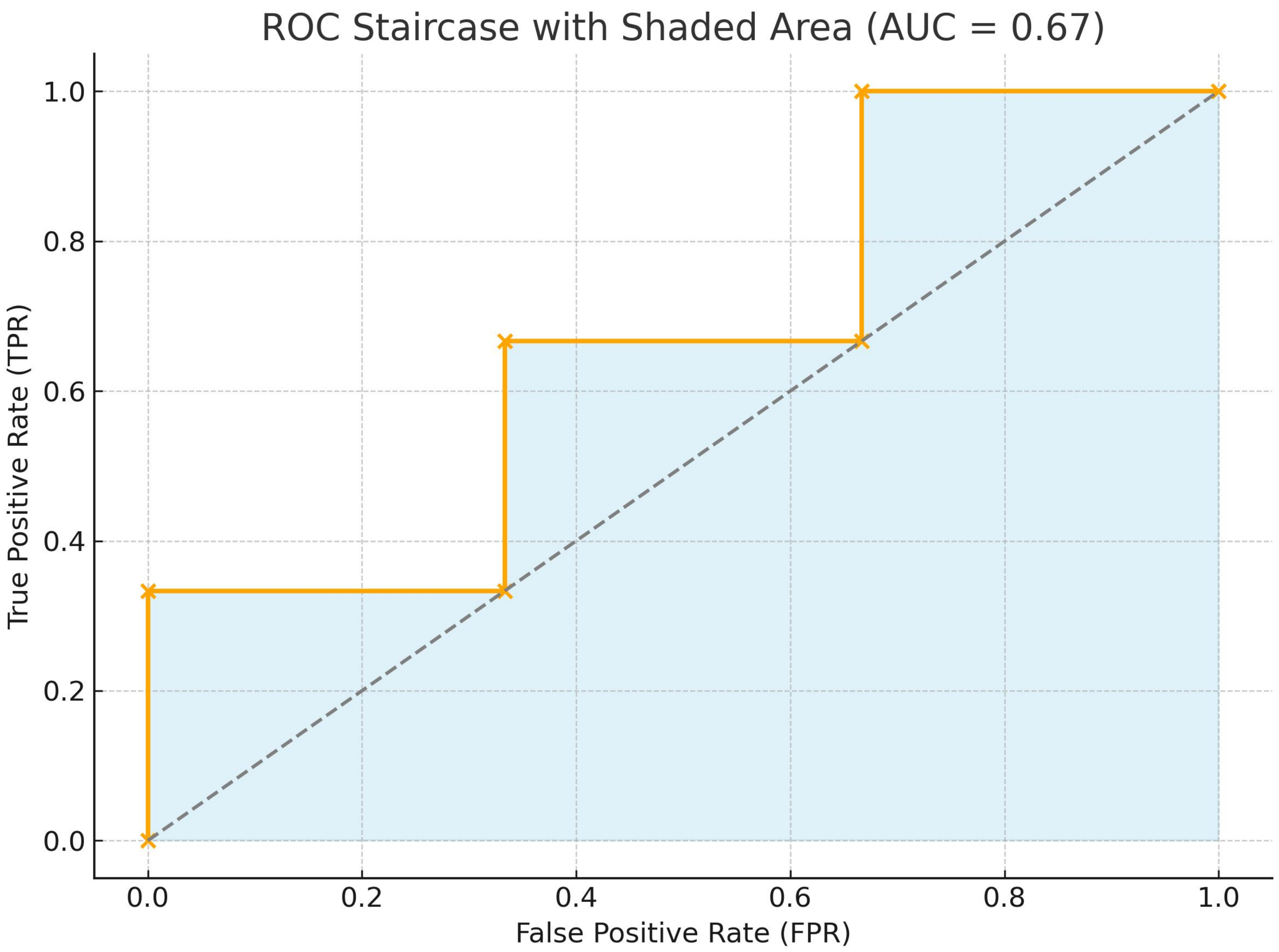
Threshold	Predicted Positives	TP	FP	TPR = TP/(TP+FN)	FPR = FP/(FP+TN)
≥ 0.95	{A}	1	0	1/3 = 0.33	0/3 = 0.00
≥ 0.90	{A,B}	1	1	0.33	1/3 = 0.33
≥ 0.85	{A,B,C}	2	1	2/3 = 0.67	1/3 = 0.33
≥ 0.70	{A,B,C,D}	2	2	0.67	2/3 = 0.67
≥ 0.40	{A,B,C,D,E}	3	2	3/3 = 1.00	2/3 = 0.67
≥ 0.10	{A,B,C,D,E,F}	3	3	1.00	3/3 = 1.00

$$AUC = \sum_i (FPR_{i+1} - FPR_i) \times \frac{(TPR_{i+1} + TPR_i)}{2}$$

- AUC ≈ 0.67

From ($FPR_i \rightarrow FPR_{i+1}$)	ΔFPR	Avg TPR	Area
0.00 → 0.00	0.00	(0.00+0.33)/2 = 0.165	0.00
0.00 → 0.33	0.33	(0.33+0.33)/2 = 0.33	0.33×0.33 = 0.1089
0.33 → 0.33	0.00	(0.33+0.67)/2 = 0.50	0.00
0.33 → 0.67	0.34	(0.67+0.67)/2 = 0.67	0.34×0.67 = 0.2278
0.67 → 0.67	0.00	(0.67+1.00)/2 = 0.835	0.00
0.67 → 1.00	0.33	(1.00+1.00)/2 = 1.00	0.33×1.00 = 0.33

$$AUC \approx 0.83$$


ROC Staircase with Shaded Area (AUC = 0.67)
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0

When to use which metric??

Scenario	Best Metric
Balanced data	Accuracy
Imbalanced data (e.g., disease detection)	Precision, Recall, F1
The dataset is balanced and the model needs to be evaluated across all thresholds.	ROC-AUC
Need to minimize false positives	Precision
Need to minimize false negatives	Recall